

Schrödinger Bridge for Non-linear ROMs

Presentation of Progress

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Problem Setup and Notation

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be two probability measures (initial and terminal marginals), time rescaled to $[0, 1]$, $\gamma > 0$ the diffusion coefficient.

We seek a time-dependent density–velocity pair $(\rho_t, b_t)_{t \in [0, 1]}$ such that:

- $\rho_0 = \mu, \rho_1 = \nu$
- ρ_t evolves according to the **Fokker–Planck equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t b_t) = \gamma \Delta \rho_t$$

- The pair minimizes the **expected kinetic energy**

$$\mathcal{J}(\rho, b) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} |b_t(x)|^2 \rho_t(x) dx dt$$

This arises from Girsanov's theorem applied to the controlled diffusion

$$dX_t = b_t(X_t) dt + \sqrt{2\gamma} dW_t,$$

where the relative entropy w.r.t. Brownian motion is $\text{KL}(\mathbb{P} \parallel \mathbb{R}) = \mathcal{J}/(2\gamma)$.

The Constrained Variational Problem

The Schrödinger bridge is the solution to

$$\inf_{(\rho, b)} \mathcal{J}(\rho, b) \quad \text{subject to} \quad \partial_t \rho_t + \nabla \cdot (\rho_t b_t) = \gamma \Delta \rho_t, \quad \rho_0 = \mu, \quad \rho_1 = \nu.$$

Introduce a Lagrange multiplier $\phi \in C^1([0, 1] \times \mathbb{R}^d)$ to enforce the constraint. The augmented Lagrangian reads:

$$\mathcal{L}(\rho, b, \phi) = \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} |b_t|^2 \rho_t - \phi_t (\partial_t \rho_t + \nabla \cdot (\rho_t b_t) - \gamma \Delta \rho_t) \right) dx dt.$$

Stationarity $\delta \mathcal{L} = 0$ with respect to ρ , b , and ϕ yields the Euler–Lagrange system.

Euler–Lagrange Optimality Conditions

Variation w.r.t. $b \Rightarrow$ optimal drift

$$b_t(x) = \nabla \phi_t(x).$$

Variation w.r.t. $\rho \Rightarrow$ Hamilton–Jacobi–Bellman equation (backward)

$$\partial_t \phi_t + \frac{1}{2} |\nabla \phi_t|^2 + \gamma \Delta \phi_t = 0.$$

Coupled Euler–Lagrange system

$$\begin{cases} \partial_t \phi_t + \frac{1}{2} |\nabla \phi_t|^2 + \gamma \Delta \phi_t = 0, \\ \partial_t \rho_t + \nabla \cdot (\rho_t \nabla \phi_t) = \gamma \Delta \rho_t, \\ \rho_0 = \mu, \quad \rho_1 = \nu. \end{cases}$$

Hopf–Cole Transformation (1/2) — Linearization

Backward wave function

Define η_t^* via the value function ϕ_t :

$$\eta_t^*(x) = \exp\left(\frac{\phi_t(x)}{2\gamma}\right).$$

Substituting into the HJB equation gives the **backward heat equation**:

$$\partial_t \eta_t^* = -\gamma \Delta \eta_t^*.$$

Forward wave function

Define η_t such that $\rho_t(x) = \eta_t(x) \eta_t^*(x)$. Substituting into the Fokker–Planck equation (using $b = 2\gamma \nabla \log \eta^*$) gives the **forward heat equation**:

$$\partial_t \eta_t = \gamma \Delta \eta_t.$$

Hopf–Cole Transformation (2/2) — Results

Schrödinger system (boundary conditions in η -variables)

$$\eta_0(x) \eta_0^*(x) = \mu(x), \quad \eta_1(x) \eta_1^*(x) = \nu(x).$$

Optimal drift

$$b_t(x) = 2\gamma \nabla \log \eta_t^*(x).$$

Neumann boundary conditions on bounded domains Ω

Zero-flux $J_t \cdot n = (\rho_t b_t - \gamma \nabla \rho_t) \cdot n = 0$ on $\partial\Omega$ reduces to:

$$\gamma \left(\eta_t \frac{\partial \eta_t^*}{\partial n} - \eta_t^* \frac{\partial \eta_t}{\partial n} \right) = 0 \quad \implies \quad \frac{\partial \eta_t}{\partial n} = 0, \quad \frac{\partial \eta_t^*}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

Both linear heat equations are solved with **homogeneous Neumann BCs**.

Log-Potential Formulation

Definition

Define the forward and backward log-potentials:

$$f_t(x) := \log \eta_t(x), \quad g_t(x) := \log \eta_t^*(x).$$

Density, drift, and marginal constraints

$$\rho_t(x) = e^{f_t(x) + g_t(x)}, \quad b_t(x) = 2\gamma \nabla g_t(x).$$

$$f_0 + g_0 = \log \mu, \quad f_1 + g_1 = \log \nu.$$

Remark

The heat equations for (η, η^*) translate to nonlinear parabolic equations for (f, g) that are *not* solved directly. Instead, IPFP alternates linear heat solves and logarithmic rescaling of boundary conditions.

IPFP in Log-Potential Form

Let $\mathcal{P}_t$ denote the heat semigroup generated by $\gamma\Delta$ with Neumann BCs.

IPFP iterations (Sinkhorn in log-form)

$$g^{(k+1)}(x) = \log \nu(x) - \log\left(\mathcal{P}_1[e^{f^{(k)}}](x)\right),$$

$$f^{(k+1)}(x) = \log \mu(x) - \log\left(\mathcal{P}_1[e^{g^{(k+1)}}](x)\right).$$

Recovery of the optimal flow

Once convergence is reached:

$$\rho_t(x) = \exp(f_t(x) + g_t(x)), \quad b_t(x) = 2\gamma \nabla g_t(x).$$

The pair (ρ, b) solves the Euler–Lagrange system and minimizes $\mathcal{J}(\rho, b)$.

The Heat Semigroup $\mathcal{P}_t$

$\mathcal{P}_t$ denotes the semigroup generated by $\gamma\Delta$ with Neumann BCs on Ω . On $\mathbb{R}^d$ it has an explicit kernel:

$$(\mathcal{P}_t f)(x) = \int_{\mathbb{R}^d} K_t(x, y) f(y) dy, \quad K_t(x, y) = \frac{1}{(4\pi\gamma t)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{4\gamma t}\right).$$

Key identity: Gibbs kernel $\equiv$ heat semigroup

The Sinkhorn kernel $K_{ij} = e^{-\|x_i - x_j\|^2/\varepsilon}$ with $\varepsilon = 4\gamma$ is *exactly* the heat semigroup $e^{\gamma\Delta}$ evaluated at $t = 1$. This means the same FVM solver used inside IPFP also computes the convolutional Sinkhorn kernel, which we reuse for the W_2 error metric.

On bounded domains

No closed-form kernel is available. We solve $\partial_t \varphi = \gamma\Delta\varphi$ numerically on the unstructured aerodynamic mesh using a Finite Volume Method with homogeneous Neumann boundary conditions.

Two-Point Flux Approximation (TPFA)

Flux across face e_{ij} separating cells K_i and K_j :

$$F_{ij} = -\gamma \frac{\varphi_j - \varphi_i}{\|x_j - x_i\|} |e_{ij}|.$$

Neumann BC: ghost-cell reflection $\varphi_j = \varphi_i$ on $\partial\Omega \Rightarrow$ zero flux. Semi-discrete:

$$\mathcal{L}(\varphi)_i = |K_i|^{-1} \sum_{j \sim i} F_{ij}.$$

SSP-RK2 (Heun's method)

$$\varphi^* = \varphi^n + \Delta t \mathcal{L}(\varphi^n), \quad \varphi^{n+1} = \frac{1}{2}(\varphi^n + \varphi^* + \Delta t \mathcal{L}(\varphi^*)).$$

CFL: $\Delta t \leq \text{CFL} \cdot h^2 / \gamma$, $h = 10^{\text{th}}$ -percentile cell size (avoids worst-cell control). Step count n is a compile-time static constant (`jax.lax.fori_loop` fusion).

Anderson Acceleration of IPFP

IPFP is the fixed-point iteration $x^{(k+1)} = F(x^{(k)})$, $x = \text{vec}(f, g) \in \mathbb{R}^{2N}$, with linear rate $\rho < 1$.

Anderson(m) extrapolation

Store the last m proposals $X = [x^{(k-m+1)}, \dots, x^{(k)}]$ and residuals $R = [r^{(k-m+1)}, \dots, r^{(k)}]$, $r^{(j)} = F(x^{(j)}) - x^{(j)}$. Find the convex combination minimising the residual norm:

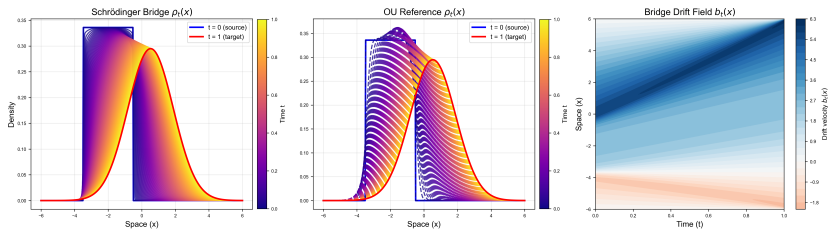
$$\min_{\substack{c \in \mathbb{R}^m \\ \mathbf{1}^T c = 1}} \|Rc\|_2^2 \implies c = \frac{(R^T R)^{-1} \mathbf{1}}{\mathbf{1}^T (R^T R)^{-1} \mathbf{1}}, \quad x^{(k+1)} = Xc.$$

Cost and convergence

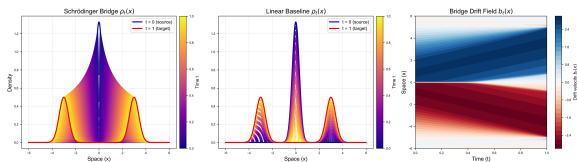
Each step: one Sinkhorn step (2 heat solves) + $\mathcal{O}(m^2 N)$ least-squares (negligible).

Anderson(5): ~ 150 – 160 measured iterations to tol 10^{-7} vs. ~ 2150 for plain Sinkhorn ($> 13\times$ faster).

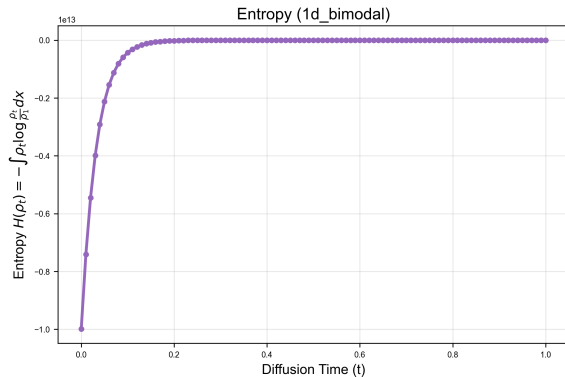
1D Validation: Square $\rightarrow$ OU Process ($\gamma = 0.05$)



1D Validation: Gaussian $\rightarrow$ Bimodal ($\gamma = 0.05$)



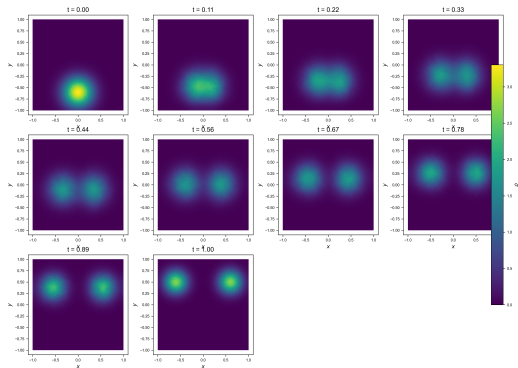
Density evolution (splitting transport)



Entropy $H[\rho_t]$ along the bridge

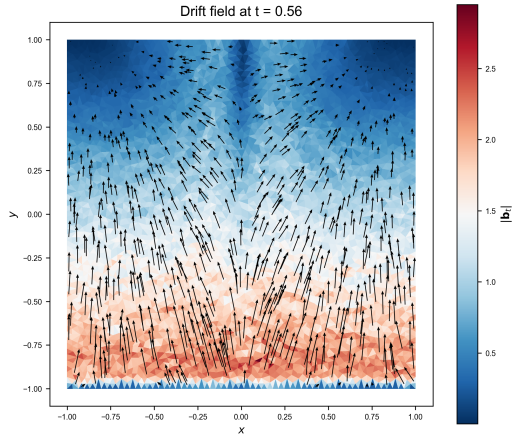
2D Validation: Gaussian $\rightarrow$ Bimodal on Unstructured Mesh

2D density transport



Unstructured density transport ($\gamma = 0.05$)

Drift field at $t = 0.56$

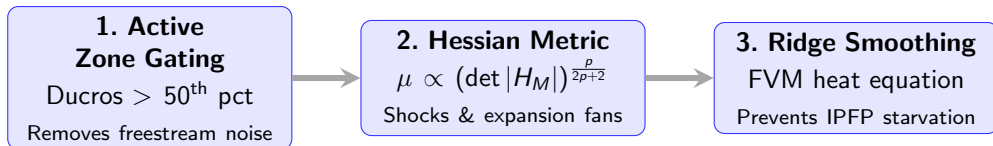


Drift field $b_t = 2\gamma \nabla g_t$ at $t = 0.5$

Application: Feature Extraction Pipeline

Problem

We need marginals μ_0, μ_1 capturing both C^0 discontinuities (shocks) and C^1 discontinuities (expansion fans) on the aerodynamic mesh.

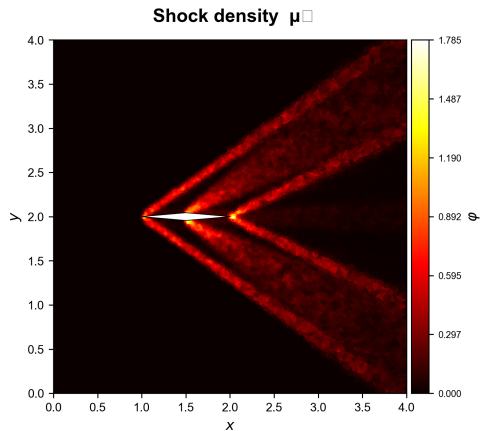


The Alauzet–Loseille metric density (two modes):

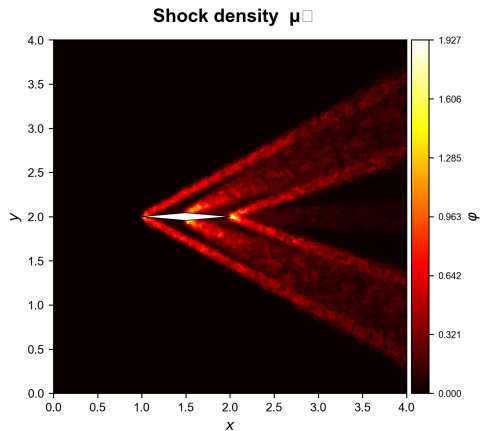
$$\det : \quad \mu(x) \propto |\lambda_1 \lambda_2|^{\frac{p}{2p+2}}, \quad \text{eig} : \quad \mu(x) \propto |\lambda_{\max}|^{\frac{p}{2p+2}},$$

where λ_1, λ_2 are eigenvalues of $|H_M|$, $p = 2$.

Extracted Marginals — Diamond Body ($M 2.00 \rightarrow M 2.50$)



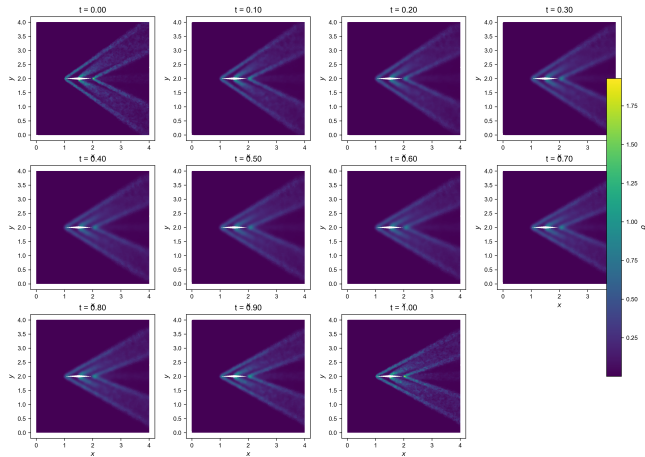
μ_0 — Mach 2.00 (det mode)



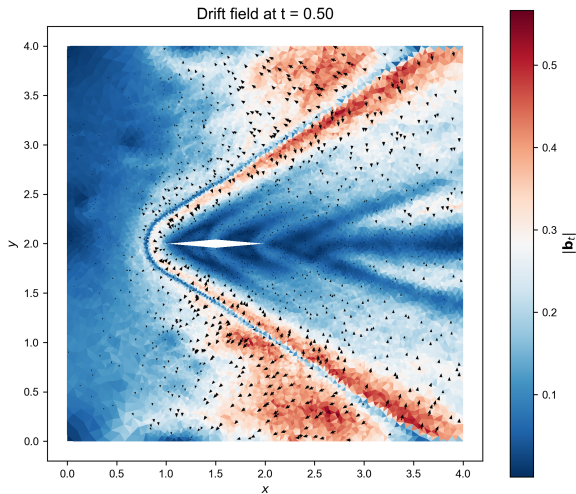
μ_1 — Mach 2.50 (det mode)

SB Density Transport — Diamond Body

2D density transport



Drift Field $b_t = 2\gamma\nabla g_t$ — Diamond Body



Barycentric SB-CDI: IPFP Potentials $\rightarrow$ Mach Interpolation (1/2)

Step 1 — IPFP $\rightarrow$ Sinkhorn potentials f, g

$$\mu_0 = e^f \cdot \mathcal{P}_1 e^g, \quad \mu_1 = e^g \cdot \mathcal{P}_1 e^f.$$

Step 2 — Barycentric transport maps (boundary-corrected conditional means)

Forward map T (from e^g) and backward map S (from e^f):

$$T(x) = \frac{\mathcal{P}_1[\text{id} \cdot e^g](x)}{\mathcal{P}_1[e^g](x)} - \beta(x), \quad S(x) = \frac{\mathcal{P}_1[\text{id} \cdot e^f](x)}{\mathcal{P}_1[e^f](x)} - \beta(x),$$

where $\beta(x) = \mathcal{P}_1[\text{id}]/\mathcal{P}_1[1] - x$ removes the Neumann identity-leak.

Step 3 — Symmetric two-map CDI reconstruction (explicit, non-iterative)

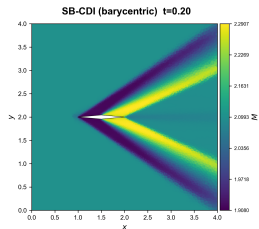
Sample each endpoint field along its own displaced pre-image, using the forward map T ($\mu_0 \rightarrow \mu_1$) and backward map S ($\mu_1 \rightarrow \mu_0$):

$$W(t, x) = (1 - t)x + tS(x), \quad Z(t, x) = tx + (1 - t)T(x),$$

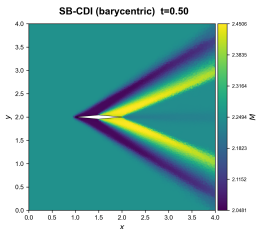
$$\hat{M}_t(x) = (1 - t)M_0(W(t, x)) + tM_1(Z(t, x)).$$

Gradient-free, no inversion, and exact at the endpoints: $\hat{M}_0 = M_0$, $\hat{M}_1 = M_1$.

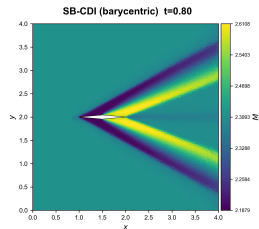
Mach Interpolation — BaryCDI vs. HR Reference (det mode)



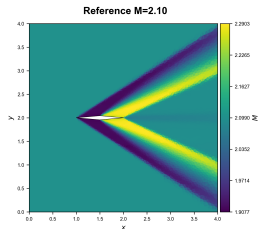
BaryCDI, $t = 0.20$



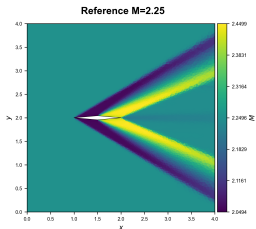
BaryCDI, $t = 0.50$



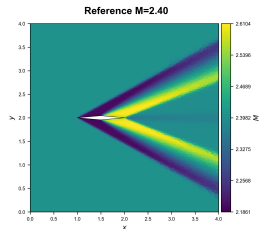
BaryCDI, $t = 0.80$



HR ref., $M = 2.10$

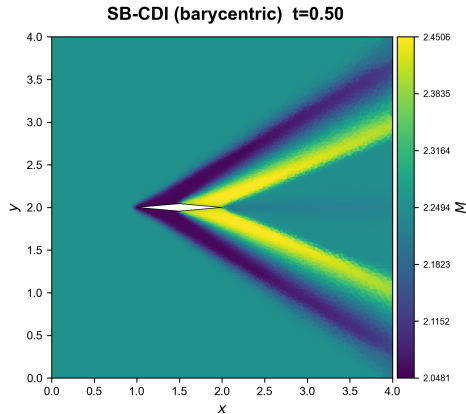


HR ref., $M = 2.25$

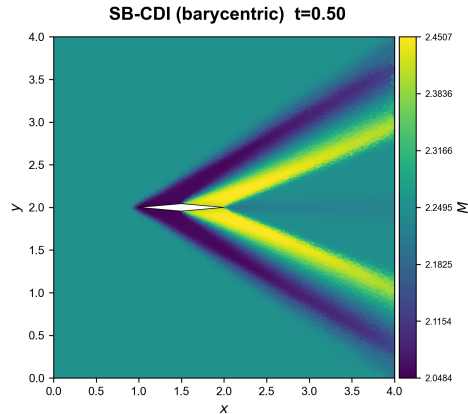


HR ref., $M = 2.40$

Mach Interpolation at $t = 0.5$ — det vs eig mode



Barycentric CDI (det mode)



Barycentric CDI (eig mode)

On the diamond's straight oblique shocks, det gives the better, more robust reconstruction; eig is preferred for curved shocks (e.g. the bump).

Quantitative Error: Wasserstein-2 Distance (1/2)

Kantorovich formulation

Given interpolated field $\hat{M}_t$ and reference M_t^{ref} , define $\mu^M(x) \propto |M(x) - M_\infty|$ (normalized to unit mass on the mesh). The W_2 distance:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{\Omega \times \Omega} \|x - y\|^2 d\pi(x, y).$$

Entropy-regularised OT cost

With regularisation parameter $\varepsilon = 4\gamma_w$:

$$\text{OT}_\varepsilon(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad C_{ij} = \|x_i - x_j\|^2.$$

Quantitative Error: Wasserstein-2 Distance (2/2)

Debiased Sinkhorn divergence

Removes the entropic self-bias so that $S_\varepsilon(\mu, \mu) = 0$:

$$S_\varepsilon(\mu, \nu) = \text{OT}_\varepsilon(\mu, \nu) - \frac{1}{2}\text{OT}_\varepsilon(\mu, \mu) - \frac{1}{2}\text{OT}_\varepsilon(\nu, \nu) \approx W_2^2(\mu, \nu).$$

Implementation via convolutional Sinkhorn (Solomon et al. 2015)

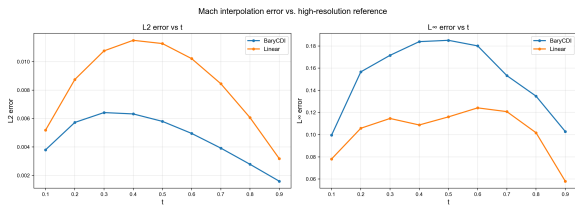
The Gibbs kernel $K_{ij} = e^{-\|x_i - x_j\|^2/\varepsilon}$ is approximated by the FVM heat semigroup $e^{\gamma_w \Delta}$, *the same operator used inside IPFP*.

Dual value at convergence: $\text{OT}_\varepsilon = \varepsilon(\langle F, \mu \rangle + \langle G, \nu \rangle)$, with Sinkhorn potentials F, G satisfying

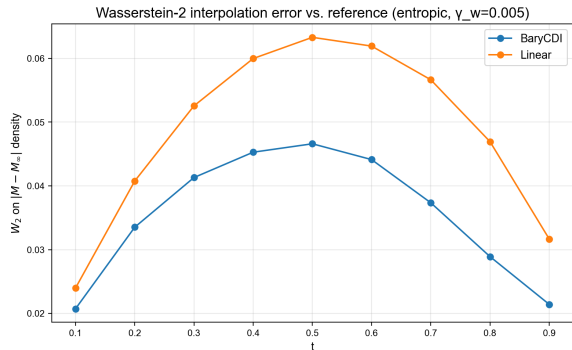
$$G = \log \nu - \log \mathcal{P}_1[e^F], \quad F = \log \mu - \log \mathcal{P}_1[e^G], \quad W_2 \approx \sqrt{\max(S_\varepsilon, 0)}.$$

Parameters: $\gamma_w = 0.005$ ($\sigma_w \approx 0.10$), 50 Sinkhorn iterations.

Error vs. High-Resolution Reference — Diamond (det mode)

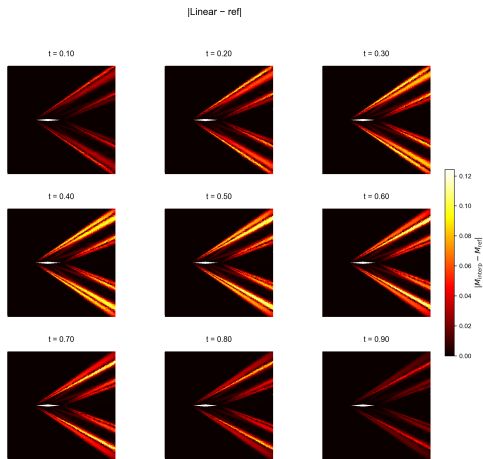


L_2 and L_∞ Mach errors vs. 9 references

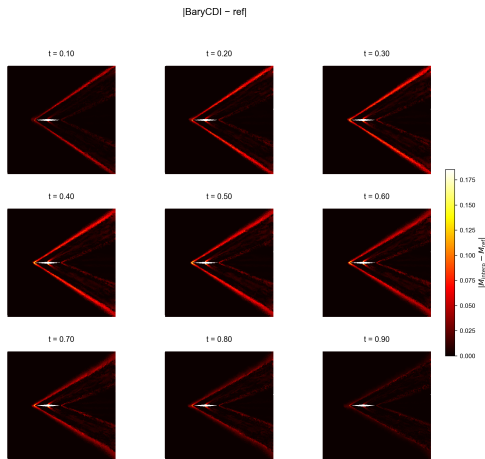


W_2 distance on $|M - M_\infty|$ density

Absolute Error Heatmaps — Diamond (det mode)



Linear interpolation



Barycentric CDI (det)